

Fuzzy Dark Matter Dynamical Friction:

stalling of globular clusters induced by dynamical heating

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Szpilfidel, Boldrini, Bovy, Di Matteo +25; submitted to A&A, arxiv:
2510.00927

Introduction

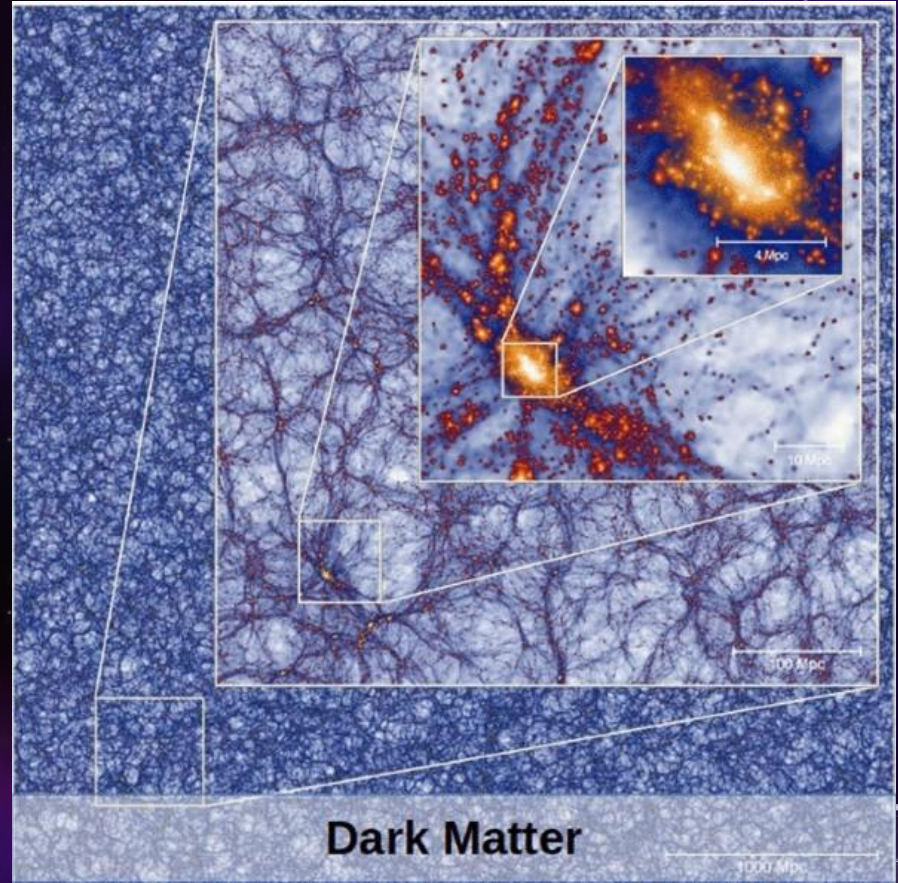
Dominant cosmological model: **Cold Dark Matter (CDM)**, describes very well the **large-scale structures** of the Universe ...

... but there are **tensions** with observations on smaller scales: **galactic scales** (1-100 kpc):

- Fornax globular cluster problem
- Cusp-core problem
- Rotation curve diversity problem
- Satellite plane
- ...

How to address these problems ?

- Baryonic processes
- Build **alternative dark matter model**



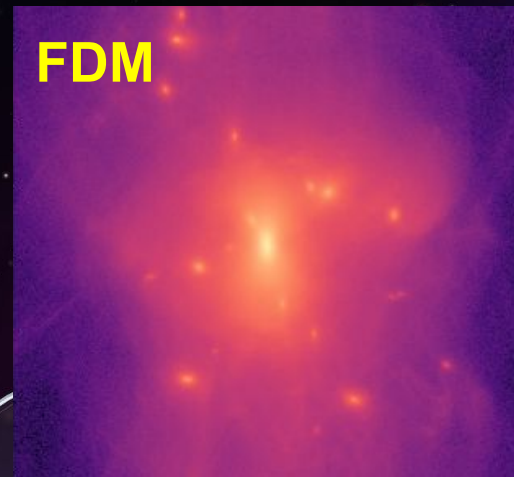
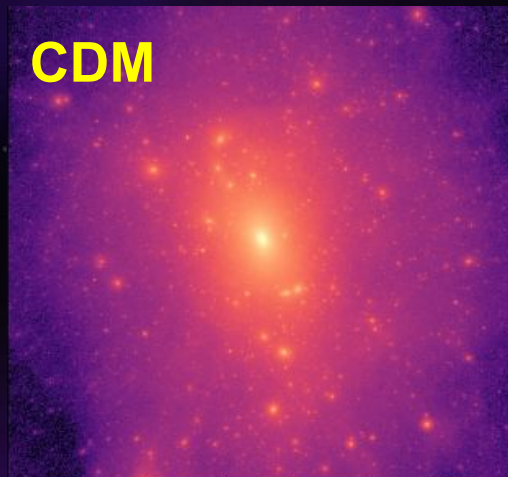
Alternative dark matter models

Several exotic dark matter (DM) models: Warm DM (*Bose et al., +05*),
Self-Interacting DM (*Spergel et al., 2000*), ...

Recent alternative: Fuzzy Dark Matter (FDM) (*Hu et al., 2000*) as an
ultralight scalar field without interaction. Constituted of a boson with mass

$$m_\chi \sim 10^{-22} \text{ eV}$$

Modifies the dynamics
on galactic scales and
converges to CDM on
larger scales.

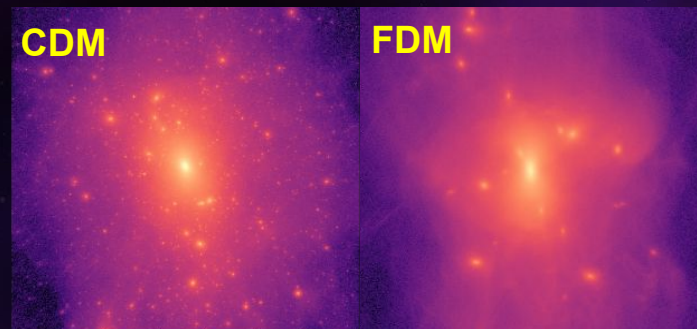


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Fuzzy Dark Matter (FDM)

Free parameter: $m_{22} = \frac{m_\chi}{10^{-22} \text{ eV}}$

Quantum wave-like nature of this particle **perturbs the density distribution**: stochastic density fluctuations.



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3 main effects on galaxies:

- Suppression of low-mass subhalo formation (*Marsh & Silk, +14*)
- Formation of a **dark matter core** in galaxies (*Schive et al, +14*)
- Reduction/suppression of **dynamical friction** (*Hui et al, +17*)

The magnitude of these effects depends on the boson mass.

$$m_{22} = 0.1$$

→ **Strong effect**

$$m_{22} = 100$$

→ **Same as CDM**

How to disentangle dark matter models ?

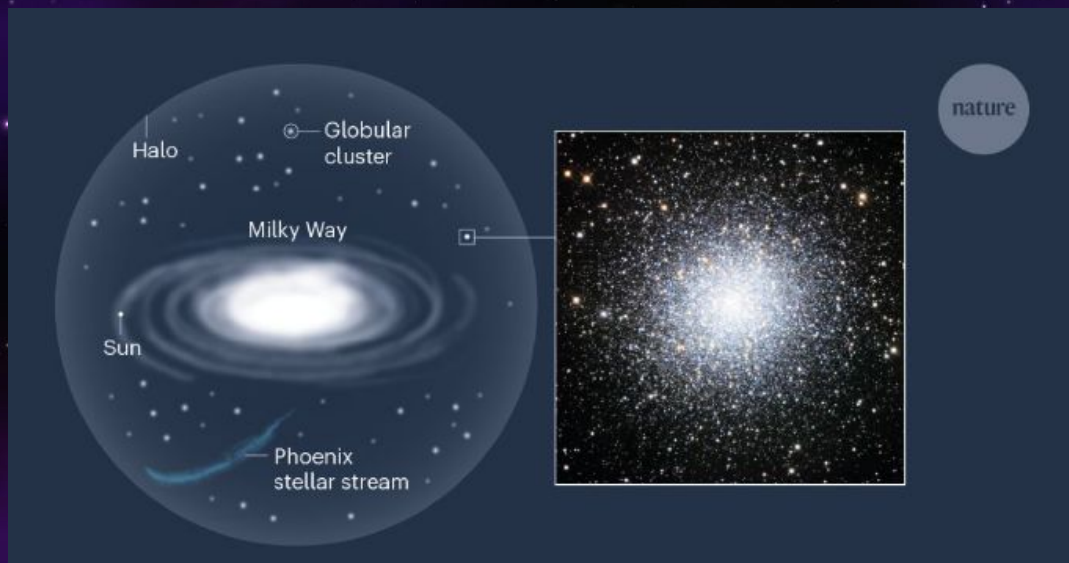
Globular clusters (GC) as tracers of DM: dense object composed of millions of stars. Their **dynamics** is sensitive to the **DM background** medium.

Mass: $M_{GC} \sim 10^4 - 10^6 M_{\odot}$

Half-mass radius: $r_{hm} \sim 10 \text{ pc}$

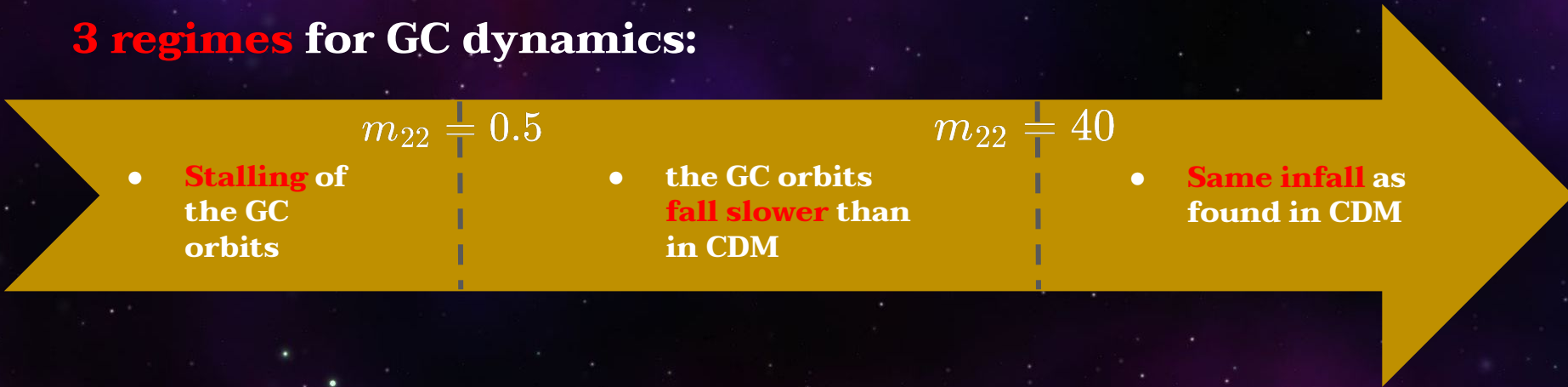
Distributed at all radii: 1-100 kpc

for the MW



Main result

3 regimes for GC dynamics:

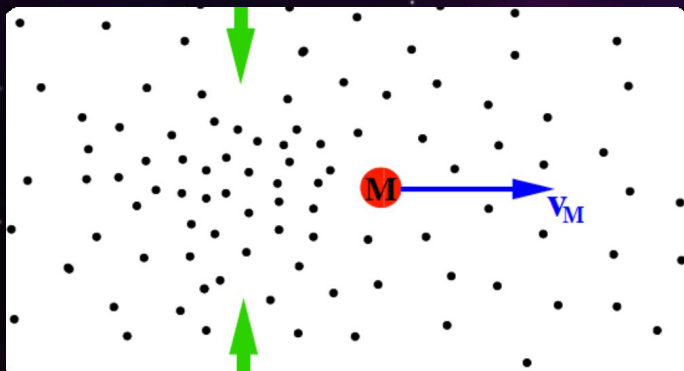
- 
- **Stalling** of the GC orbits
 - the GC orbits **fall slower** than in CDM
 - **Same infall** as found in CDM

➤ The regimes depend on the value of the **free parameter**

$$m_{22} = \frac{m_\chi}{10^{-22} \text{ eV}}$$

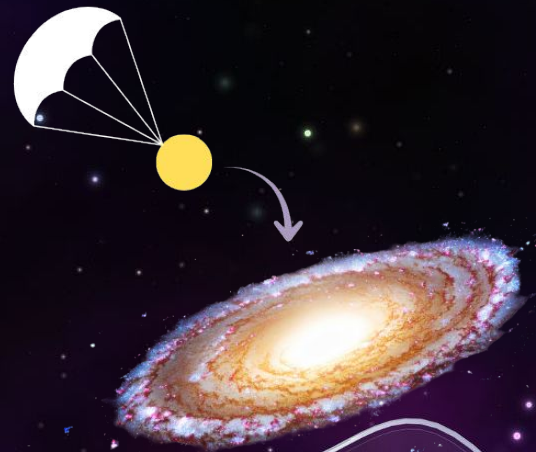
Dynamical friction as a probe of dark matter nature

Dynamical friction: **Energy loss** mechanism, drives the **infall of massive objects** (black holes, globular clusters, galaxy satellites).



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$$\mathbf{F}_{\text{DF}} \propto - \frac{M_{\text{GC}}^2 \rho_{\text{DM}}(r)}{v^3} \mathbf{v}$$



➤ **Looking at the infall of GC provides insight on the DM density profile**

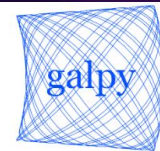
FDM dynamical friction formalism

First implementation of the FDM formalism, **new public instance available in GALPY**

```
class galpy.potential.FMDynamicalFrictionForce(amp=1.0, GMs=0.1, gamma=1.0, rhm=0.0,
m=1e-99, dens=None, sigmar=None, const_lnLambda=False, const_FDMfactor=False, minnr=0.0001,
maxxr=25.0, nr=501, ro=None, vo=None)
```

[\[source\]](#)

Implements the fuzzy dark matter (FDM) dynamical friction force.



$$\mathbf{F}_{\text{DF}}(r) = -4\pi G^2 M_{\text{GC}}^2 \rho(r) \frac{\mathbf{v}}{v^3} \mathcal{C}(r, v)$$

$$\mathcal{C}_{\text{CDM}}(r, v) = \frac{1}{2} \ln(1 + \Lambda^2) \left[\text{erf}\left(\frac{v}{\sqrt{2}\sigma}\right) - \frac{2v}{\sqrt{2\pi}\sigma} \exp\left(-\frac{v^2}{2\sigma^2}\right) \right]$$

CDM (*Chandrasekhar, 1943; Petts et al. +15*)

2 physical regimes in FDM:

- ❖ Zero velocity dispersion in the medium (*Hui et al. +17*)
- ❖ Velocity dispersion in the medium (*Lancaster et al. +20*)

$$m_{22} = 0.1$$

$$C_{\text{FDM}} \ll C_{\text{CDM}}$$

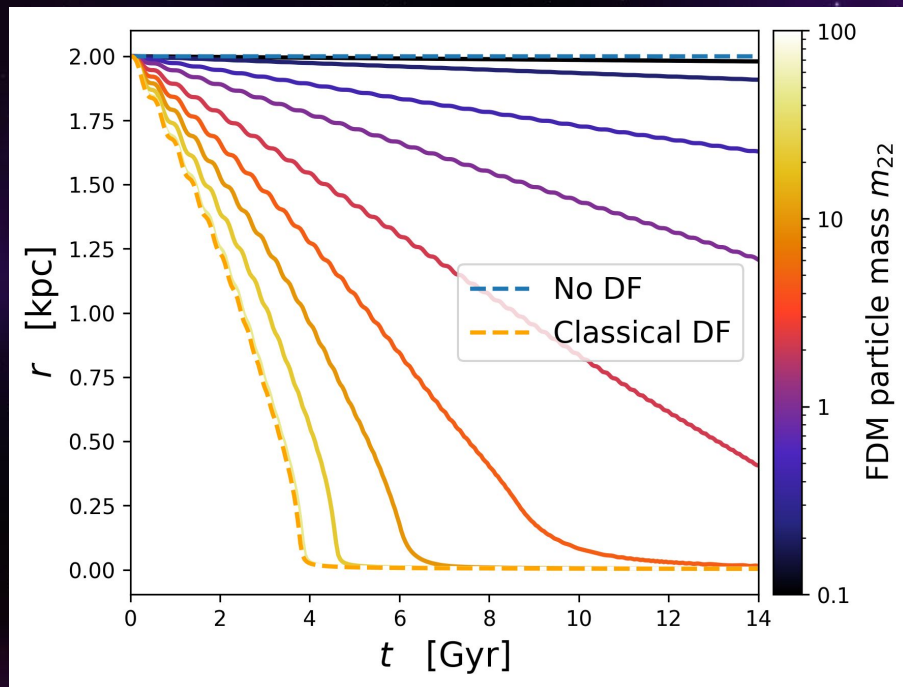
$$m_{22} = 100$$

$$C_{\text{FDM}} \rightarrow C_{\text{CDM}}$$

Dynamical friction in FDM

- Stochastic density fluctuations: granules.
- Density fluctuations in FDM **increase kinetic energy** through **dynamical heating**.
- **Dynamical heating** counteracts **dynamical friction**.

$$m_{22} = \frac{m_\chi}{10^{-22} \text{ eV}}$$



$$m_{22} = 0.1$$

- More dynamical heating
- Longer fall-in time

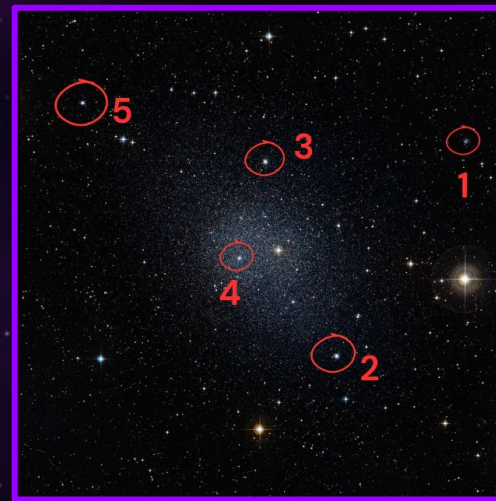
$$m_{22} = 100$$

- Less dynamical heating
- Shorter fall-in time (same as CDM)

Fornax GC problem

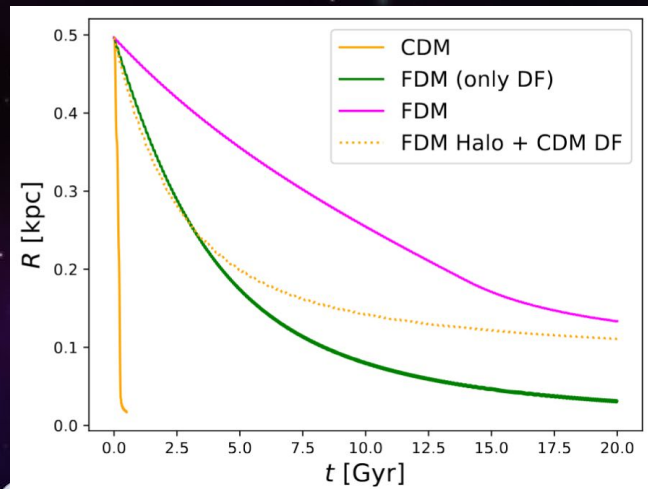
Fall-in time of GCs	CDM (Gyr)	FDM $m_{22}=3$ (Gyr)	FDM $m_{22}=20$ (Gyr)
Fornax 1	25	393	176
Fornax 2	2.56	59	26
Fornax 3	0.3	19	11
Fornax 4	0.24	24	13
Fornax 5	5	74	34

FDM provides an explanation for the survival of Fornax's GCs



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Fornax 3 orbit

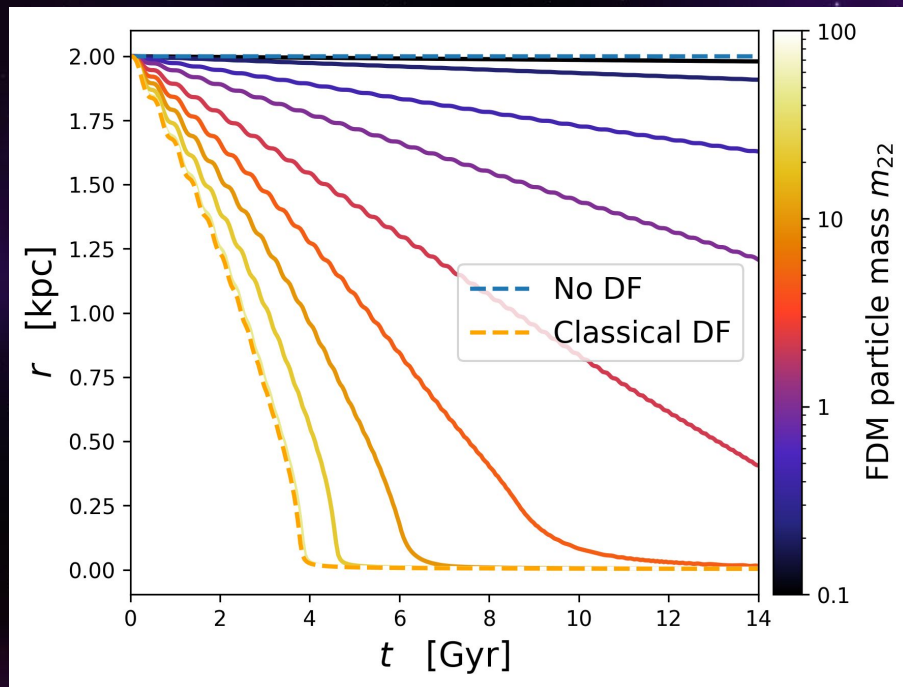


Generalization of the analysis

Analysis of the **effect of FDM** on various **GC orbits** in DM halos:

Orbital integrations:

- 1000 GCs with mass $10^6 M_{\text{sol}}$, distributed in all the E- L_z space.
- DM halo masses: $M_{\text{halo}} = 10^9 - 10^{12} M_{\text{sol}}$
- FDM boson mass $m_{22} = 0.1 - 100$



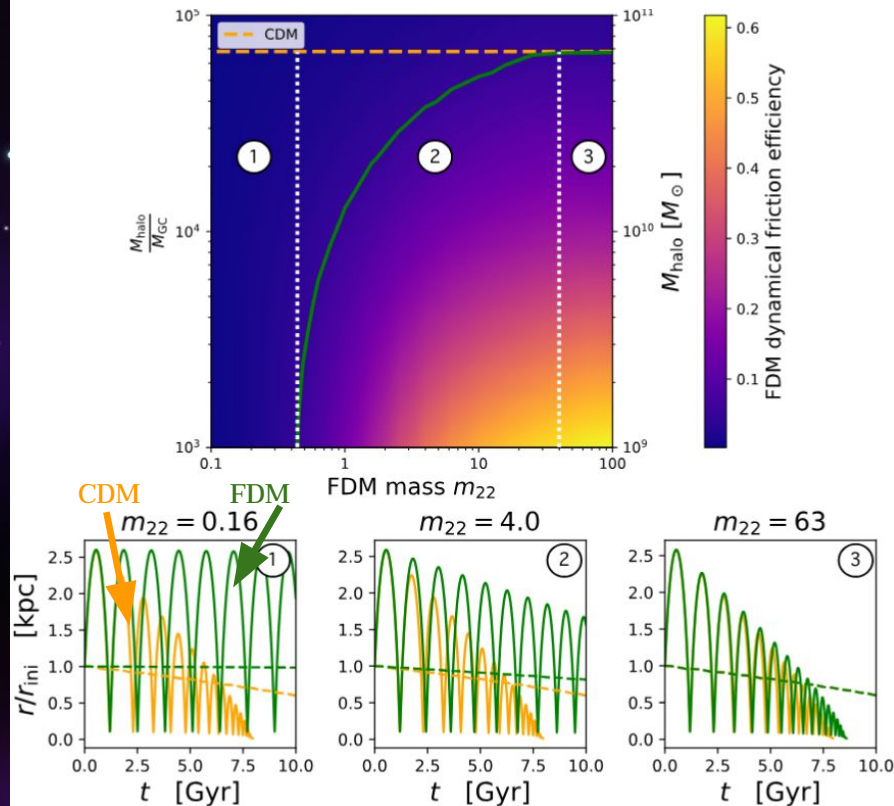
Dynamical friction in FDM

$$\text{DF efficiency} = \frac{|r_{\text{apo}}^{\text{DF}} - r_{\text{apo}}^{\text{no DF}}|}{r_{\text{apo}}^{\text{no DF}}} > 5\%$$

Classical DF: inefficient for mass ratio $> 7 \times 10^4$

FDM DF: 3 regimes for GC dynamics

NFW + FDM DF



$m_{22} = 0.1$

$m_{22} = 0.5$

$m_{22} = 40$

$m_{22} = 100$

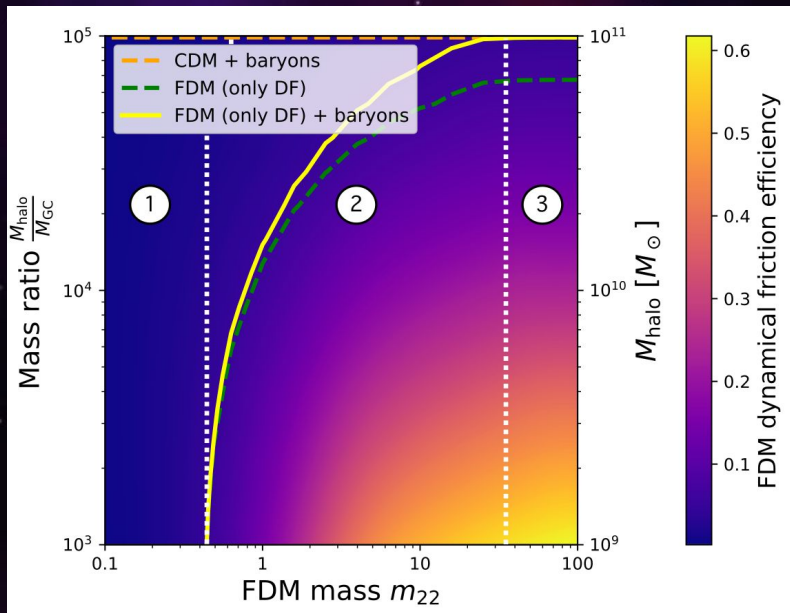
➤ DF completely inefficient

➤ DF significantly reduced

➤ FDM DF similar than in CDM

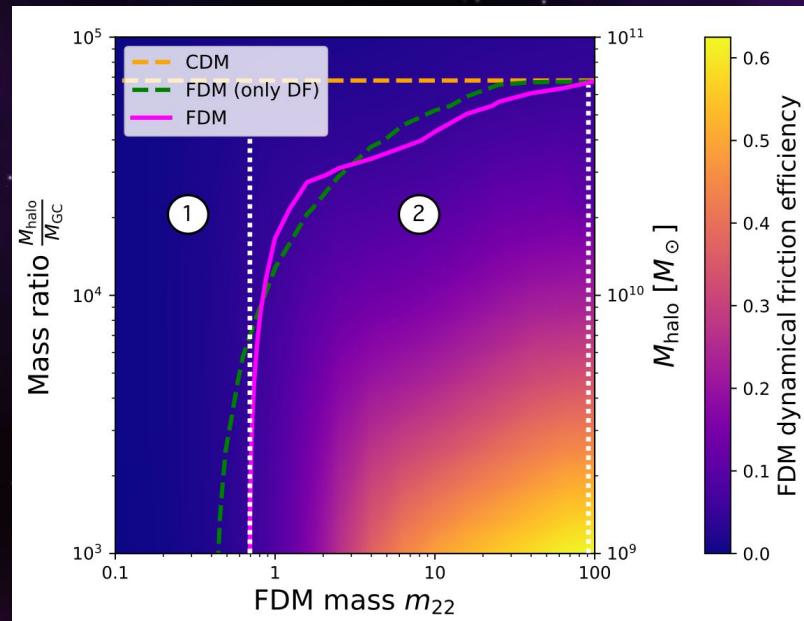
More complete modelisation

NFW + FDM DF + Stellar component



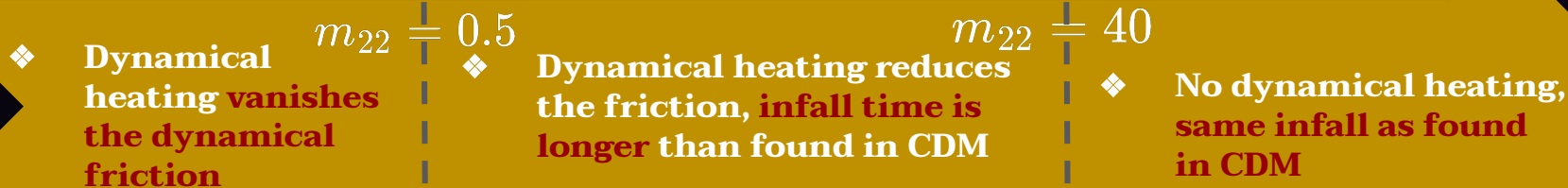
DF overall **more efficient** (yellow curve): baryons **increase the density**
Reduced DF regime slightly tightened.

DM core + FDM DF



DM core **increase the DF inefficiency**
Broadening of the reduced DF regime.

3 regimes for FDM dynamical friction:

- 
- ❖ Dynamical heating **vanishes** the dynamical friction $m_{22} = 0.5$
 - ❖ Dynamical heating reduces the friction, **infall time is longer** than found in CDM $m_{22} = 40$
 - ❖ No dynamical heating, **same infall as found in CDM**

Cosmological constraints: $m_{22} = 7 - 20$ (Lya forest) (*Irsic +17, Kobayashi +17*)

Other galactic constraints: $m_{22} = 0.5 - 1.5$ (*Amorisco+18a, Chiang+23b*)

Perspectives

- New numerical package, useful for **GC dynamics** in FDM in other galaxies.
- Euclid: Look for other dwarf galaxies with **timing problem** and apply FDM formalism.

Thank you!